Customer Churn Prediction
Analysis and Conclusions

Model: K-Nearest Neighbors (KNN) **Dataset:** Telco Customer Churn

1. Dataset and Approach

The Telco Customer Churn dataset contains 7,043 customers and 21 original columns. There are 5,174 customers who did not churn and 1,869 customers who churned, meaning approximately 26.5% of the customers churned.

Before training the model, the customer ID was removed, TotalCharges was converted to a numeric variable, and the 11 missing values created during conversion were handled using the median. The target variable Churn was converted from Yes/No to 1/0, and the categorical variables were encoded. After encoding, the model contained 30 input features.

The data was divided into 80% training data (5,634 customers) and 20% testing data (1,409 customers). Stratification was used so that the proportion of churn and non-churn customers remained similar in both datasets.

2. Initial KNN Model (K = 5)

The first KNN model was trained using five nearest neighbors. Its performance was:

Metric	Result
Training Accuracy	82.77%
Test Accuracy	76.58%
Precision	57.91%
Recall	43.05%

The confusion matrix showed that 918 non-churning customers were correctly identified, 117 non-churning customers were incorrectly predicted to churn, 161 churners were correctly identified, and 213 actual churners were missed. This shows that the model performs better at identifying customers who will stay than customers who will churn.

3. Experimenting with Different K Values

Different K values were tested to determine how the number of neighbors affects model performance.

K	Accuracy	Precision	Recall
1	71.19%	45.70%	45.45%
3	76.22%	56.57%	44.92%
5	76.58%	57.91%	43.05%
7	78.14%	62.60%	43.85%
9	78.85%	65.32%	43.32%
11	78.64%	65.94%	40.37%
15	78.85%	67.12%	39.84%

K = 9 and K = 15 achieved the same highest accuracy of 78.85%. The program selected K = 9 because it is the first K value with the maximum accuracy. K = 9 is also preferable between these two results for churn detection because its recall (43.32%) is higher than the recall for K = 15 (39.84%).

4. Interpretation of the Final Model

The selected K = 9 model achieved 78.85% accuracy, 65.32% precision, and 43.32% recall. The accuracy means that the model predicts the correct churn outcome for approximately 79 out of every 100 customers. The precision of 65.32% means that when the model predicts that a customer will churn, it is correct about 65% of the time.

The recall of 43.32% is an important limitation. It means that the model detects only about 43% of the customers who actually churn. Therefore, more than half of the actual churners may not be identified by the model.

5. Business Recommendations

The company could use this model as an initial churn-risk screening tool. Customers predicted to be at risk of churn could be targeted with retention measures such as proactive customer support, service improvements, or targeted offers.

However, the KNN model should not be used alone for important retention decisions because its recall is relatively low. Missing a large proportion of actual churners reduces the company's ability to intervene before those customers leave.

6. Limitations and Future Improvements

A major limitation is the model's relatively low recall for the churn class. The dataset is also imbalanced: approximately 73.5% of customers did not churn and 26.5% did churn. Therefore, overall accuracy alone does not fully describe how well the model identifies customers who are likely to leave.

Future improvements could include scaling the numerical features, investigating methods for handling class imbalance, improving feature selection, and comparing KNN with other classification algorithms. The main goal of further improvement should be to increase recall while maintaining acceptable precision and accuracy.

7. Conclusion

The KNN model achieved a best accuracy of 78.85% with K = 9, with 65.32% precision and 43.32% recall. The model performs reasonably well overall but is considerably better at identifying customers who will remain than customers who will churn. Since detecting potential churners is important for customer retention, future improvements should focus on increasing recall and reducing the number of churners that the model fails to identify.